

Y-88 Energy Calibration, Gain Curves, and Trigger Thresholds

n_TOF EAR2 X17 — Y-88 source scan (224476–79) + long run 224503

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analysis and slides prepared with Claude (Anthropic AI assistant)

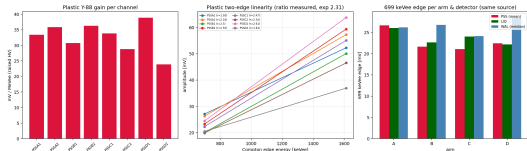
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What we did

- ▶ **Y-88 source scan** (224476–79, one source per arm): an absolute energy calibration from the **699 & 1612 keV Compton edges**, on plastics, liquids, and SiPM walls.
- ▶ Reprocessed with the LIQ PSA UserInput (the official files have no LIQ).
- ▶ Anchored the two **plastic HV scans** (224466 BNC-T, 224489 FIFO) to the Y-88 scale → absolute gain curves for setting HV.
- ▶ **Long run 224503**: SiPM MIP (per arm) and plastic triple MIP recheck.
- ▶ **Two significant discrepancies found** (flagged, need follow-up).
- ▶ Deliverable: **trigger threshold estimates** for SiPM sums and plastics.

Scripts 21--23f; docs HV_GAIN_Y88_ANALYSIS.md, report/y88_report.pdf.

Y-88 absolute scale: both Compton edges, all three detectors



- ▶ Two edges on every plastic; measured 1612/699 ratio ≈ 2.4 (expect 2.31) *confirms* the energy assignment.
- ▶ Plastic gain **35–64 mV/MeVee** (HV raised for these runs).
- ▶ **First LIQ scale:** 32–37 mV/MeVee, consistent across arms.
- ▶ At 699 keVee all three detectors land within 21–27 mV per arm.

Discrepancy #1: plastic MIP is mis-calibrated by $\sim 40\times$

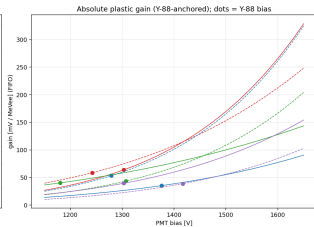
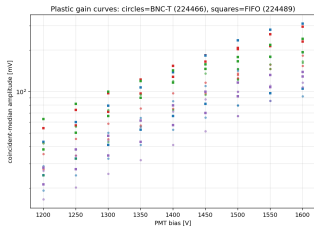
- ▶ The Y-88 line (known 699 keVee) gives plastic gain **35–64 mV/MeVee**.
- ▶ The old triples “MIP” (19d) assumed a **5.05 MeV** deposit and got 0.65–2.05 mV/MeV.
- ▶ $\Rightarrow$ the triples “MIP” peak really sits at $\sim$ **130 keVee**, not 5 MeV — a $\sim$ **40 $\times$** disagreement.
- ▶ Y-88 (BNC-T) and the 224489 MIP (FIFO) are otherwise *consistent* once the per-PMT FIFO factor (1.13–1.65) and HV are applied.

A real MIP in 2 cm plastic = 3.4 MeV (Landau MPV) $\rightarrow$ **120–217 mV** on the Y-88 scale.

The triples never find it (next slides).

Trust Y-88; recheck the triples MIP with a dedicated low-rate / prompt-beam run.

Gain curves + Y-88 anchor $\rightarrow$ setting the HV

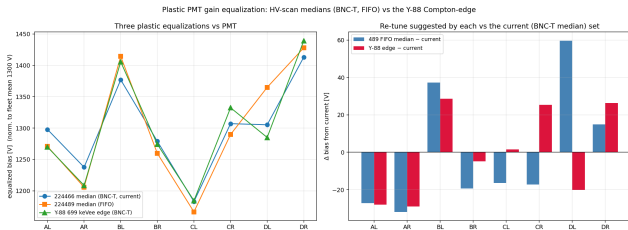


- ▶ Circles = BNC-T (224466), squares = FIFO (224489); differ by the per-PMT FIFO factor only.
- ▶ Y-88 pins the absolute mV/MeVee vs bias (right; dots = current bias).
- ▶ **Equalize + scale:** per-PMT bias to put 699 keVee at a chosen mV (table in HV_GAIN_Y88_ANALYSIS.md); e.g. 50 mV target $\Rightarrow$ AL 1324 ... DR 1559 V.

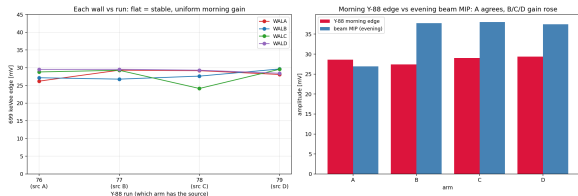
Do the three plastic equalizations agree?

Three equalizations, all normalized to fleet-mean 1300 V:

- ▶ 224466 median (BNC-T) = **current fixed voltages**;
- ▶ 224489 median (FIFO); Y-88 699 keVee edge (BNC-T).
- ▶ **Same broad pattern** (CL lowest, DR highest); agree to $\sim \pm 25$ **V RMS**.
- ▶ Y-88 edge vs current: RMS **23 V**; FIFO median vs current: 31 V.
- ▶ Mostly they point the same way (AL/AR down, BL/DR up)...
- ▶ ... but **DL and CR disagree**: the MIP-tagged median and the 0.7 MeV edge want *opposite*

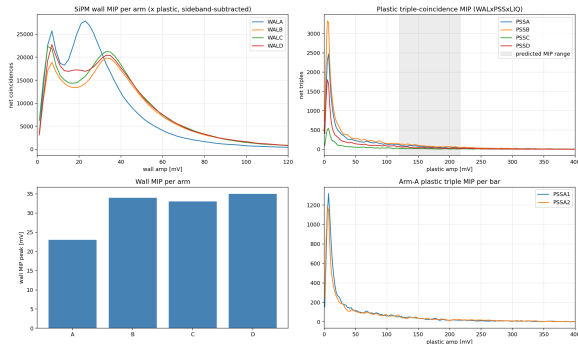


Discrepancy #2: SiPM MIP differs per arm (A low)



- ▶ Y-88 wall edge is **uniform & stable** across all 4 morning runs (~ 28 mV, left).
- ▶ But the beam MIP is **arm-A-low**: $A \approx \text{MIP}$, $B/C/D \sim 1.4\times$ higher (right).
- ▶ **Persists** in 224503 (next slide) $\Rightarrow$ not a transient.
- ▶ Uniform Y-88 (isotropic source) + non-uniform MIP (beam) $\Rightarrow$ **geometric** (per-arm beam incidence angle $\rightarrow$ path length), not a gain problem. *Confirm vs survey angles.*

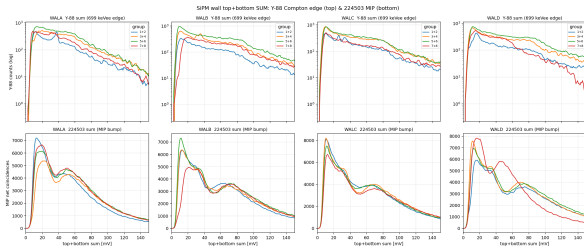
Long run 224503: SiPM MIP clear, plastic MIP swamped



- ▶ **SiPM wall MIP** (top-left): clean bumps, WALA ~ 23 , B/C/D $\sim 33\text{--}35$ mV — the arm-A gap persists.
- ▶ **Plastic triples** (top-right): net peaks at $\sim 5\text{--}7$ mV, and is *flat* across the predicted 120–217 mV MIP band.
- ▶ Not statistics — the plastic's huge low-energy singles rate ($\sim 20\text{k}$ hits/bunch) **swamps** the triple, and the late-tof cut drops prompt beam MIPs.
- ▶ $\Rightarrow$ triples can't pull the plastic MIP from a high-rate beam run.

SiPM trigger thresholds (top+bottom SUM, $0.5 \times \text{MIP}$)

Trigger fires on the top+bottom **sum** of each bar group
[(1+2)(3+4)(5+6)(7+8)]. MIP bump of the sum (bottom row), threshold = half:



wall	MIP-sum [mV]	thr [mV]
WALA	50	25
WALB	70	35
WALC	67	34
WALD	72	36

Arm A lower (Discrepancy #2). Per-channel values in
`calib/sipm_sum_thresholds.json`.

Plastic trigger thresholds (Y-88-anchored MIP estimate)

No usable triple MIP $\Rightarrow$ predict the plastic MIP from the **Y-88 scale**: $\text{MIP} = 3.4 \text{ MeV (2 cm PVT)} \times \text{gain}$.

PMT	AL	AR	BL	BR	CL	CR	DL	DR
predicted MIP [mV]	217	200	120	181	137	148	134	132
thr @1.7 MeV [mV]	109	100	60	91	68	74	67	66

- ▶ Threshold shown at 1.7 MeV ($\approx$ half the MIP); scales linearly — pick your energy.
- ▶ Per-channel because the current bias isn't perfectly equalized; equalize first (gain-curve slide) for a single common threshold ($\sim 122 \text{ mV}$ at a $50 \text{ mV}/699\text{-keV}$ target).
- ▶ Plastic full scale $\sim 2000 \text{ mV}$, so huge headroom; threshold well clear of the $\sim 5 \text{ mV}$ noise.

Recommended plastic HV: Y-88-edge equalization (FIFO)

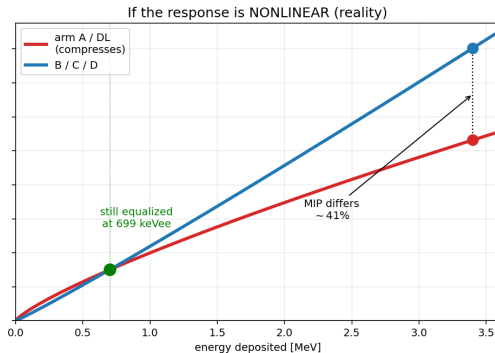
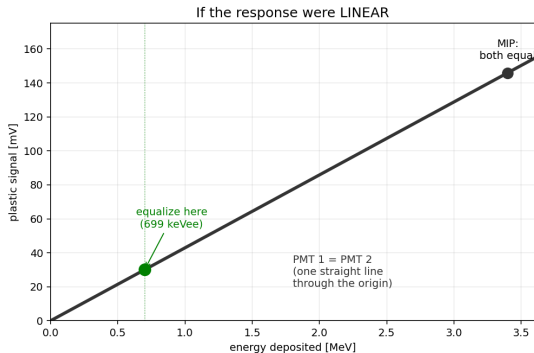
Equalize each PMT's **699 keVee edge** (slid along its own power law) to a common target; target chosen to keep the **fleet-mean HV at 1301 V** (overall gain unchanged). Y-88 is BNC-T, so folded in the per-PMT FIFO factor.

	AL	AR	BL	BR	CL	CR	DL	DR
CAEN 07...	.000	.001	.002	.003	.004	.005	.006	.007
current V	1303	1242	1376	1279	1180	1307	1303	1417
set V	1237	1177	1440	1248	1214	1312	1331	1448
change	-66	-65	+64	-31	+34	+5	+28	+31

- ▶ Arm A comes *down* (~ 65 V): it has the highest FIFO factor (1.65/1.56), so it over-gains in FIFO. BL/DL/DR/CL come up.
- ▶ Range 1177–1448 V (HV scans ran to 1600 V fine).
- ▶ Machine-readable: `calib/plastic_hv_equalized_y88.json`. Target is a scale knob — raise it to shift all gains up together.

The one caveat, explained: response **nonlinearity**

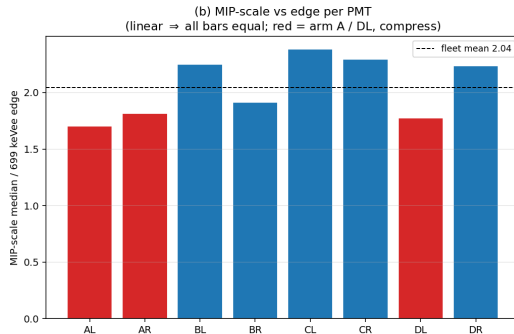
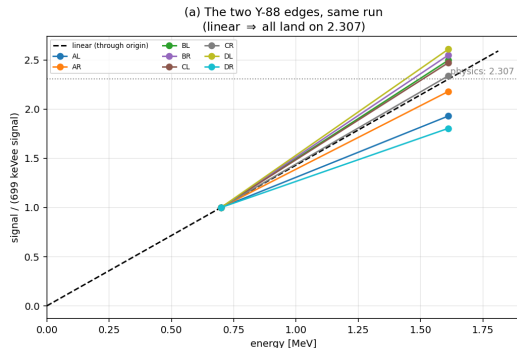
Equalizing at ONE energy only equalizes ALL energies if the response is linear



- ▶ We equalize so every PMT's **699 keVee edge** reads the same mV.
- ▶ **If the response were linear** (signal $\propto$ energy, through the origin): matching at 699 keVee makes the whole response curves identical, so the MIP is *automatically* equal too — equalize once, done (left).
- ▶ **If it is nonlinear** (the curve bends, and by a different amount per PMT): the curves cross at 699 keVee but *diverge* at higher energy, so the MIP is *not* equal even though the edge is (right).

The nonlinearity, in our data

Plastic response nonlinearity in the data: the high-energy / low-energy signal ratio varies per PMT



- ▶ **(a) Same-run check:** the two Y-88 edges (699 & 1612 keVee). If linear, $\text{signal}(1612)/\text{signal}(699) = 2.307$ for every PMT; measured 1.8–2.6 (the 1612 edge is the weaker fit, so read the per-PMT values loosely — but the spread is real).
- ▶ **(b) Robust, higher lever:** MIP-scale median / 699 keVee edge. Linear $\Rightarrow$ one constant for all 8 PMTs; measured **1.70–2.38** (1.4 $\times$). Arm A / DL sit $\sim 15\text{--}20\%$ low — those tubes **compress** more at high energy.

How to deal with the nonlinearity — suggestions

The effect is **small** ($\sim \pm 12\%$ **at the MIP**). Options, in order of effort:

1. **Accept it (recommend now).** Equalize on the Y-88 edge (absolute, known energy). MIP/trigger then uniform to $\sim \pm 12\%$. Set a common 0.5-MIP threshold ~ 76 mV; per tube it is 63–88 mV.
2. **Per-PMT thresholds** (small bookkeeping): keep the Y-88 equalization but set each tube's threshold to its own 0.5-MIP (**63–88 mV**) $\Rightarrow$ the *trigger point* is uniform even though the gain isn't, exactly.
3. **Equalize at the trigger energy** instead of the edge: use the MIP/median scale (~ 1.7 MeV, closest to the trigger). Uniform trigger, but you lose the absolute anchor and re-inherit the median's caveats.
4. **Characterize & correct the full curve:** a multi-point calibration (more source lines / a dedicated MIP+edge study) per PMT. Only worth it if a physics analysis needs $< 5\%$ linearity everywhere.

My recommendation: (1) now — the $\pm 12\%$ is small and keeps the absolute scale; add (2) if the trigger uniformity matters more than the absolute energy scale. (4) only on physics demand.

Summary & follow-ups

Thresholds to set now:

- ▶ SiPM sums: WALA **25**, WALB **35**, WALC **34**, WALD **36** mV (half the MIP-sum).
- ▶ Plastics: **60–110 mV** per PMT (1.7 MeV, Y-88 scale).

Two discrepancies to follow up:

1. Plastic triples “MIP” is at ~ 130 keVee, $\sim 40\times$ below a real 3.4 MeV MIP — the old `pss_mip_calib` energy scale is wrong; trust Y-88. Recheck triples with a low-rate / prompt-beam run.
2. SiPM MIP is $\sim 1.4\times$ lower on arm A than B/C/D, persistently — likely geometric (beam angle). Confirm against the survey.

Figures `figures/21_y88/`; calibs `calib/y88/*.json`, `calib/sipm_sum_thresholds.json`, `calib/plastic_hv_gain_absolute.json`.